



Southeast University

Department of Computer Science and Engineering

LAB Report

Course Name: Computer Networking LAB

Course Code: CSE 342.2

Experiment No.: 05

Experiment Name: Inter-vlan Configuration

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Batch: 68

Experiment NO: 05

Experiment Name: Inter-vlan Configuration [Router-on-a-stick]

Objective:

- To Implement and verify Inter-vlan routing using a Router-on-a-stick topology.
- To establish 802.1Q trunking on designated switch interfaces to carry multiple vlan traffic.
- To segment broadcast domains by configuring vlan 10 and vlan 20 and assigning access ports appropriately.
- To utilize out-of-band management via console connections for initial device configuration.

Topology Analysis: Based on the provided Physical and logical topology map, the network relies on a single Inter-vlan routing gateway (Router0) connected to the Primary distribution switch (Switch0) via a single Physical link (Gig 0/1 on Router0 to Gig 0/1 on Switch0). Switches are trunked with each other. The end devices are logically separated into two distinct subnets mapped to vlan 10 and vlan 20. Out of band management is established using console cables from Laptop0 to Router0 and Laptop1 to Switch1.

Network Addressing Table:

Device	Interface	IP Address	Default Gateway	VLAN Assignment
Router 0	Gig 0/1.10	192.168.1.1/24	N/A	802.1Q 10
Router 0	Gig 0/1.20	192.168.2.1/24	N/A	802.1Q 20
PC-PT	Fa 0/2	192.168.1.10/24	192.168.1.1	Vlan 10
Laptop-PT	Fa 0/3	192.168.1.11/24	192.168.1.1	Vlan 10
PC-PT	Fa 0/11	192.168.2.10/24	192.168.2.1	Vlan 20
Laptop-PT	Fa 0/12	192.168.2.11/24	192.168.2.1	Vlan 20
Laptop-PT	Fa 0/1	192.168.1.12/24	192.168.1.1	Vlan 10
Laptop-PT	Fa 0/2	192.168.2.12/24	192.168.2.1	Vlan 20

Device Configuration:

• Switch 0:

Switch > Enable

Switch# configuration terminal

Switch(config)# Vlan 10

Switch(config)# Name Students

Switch(config)# Exit

Switch(config)# Vlan 20

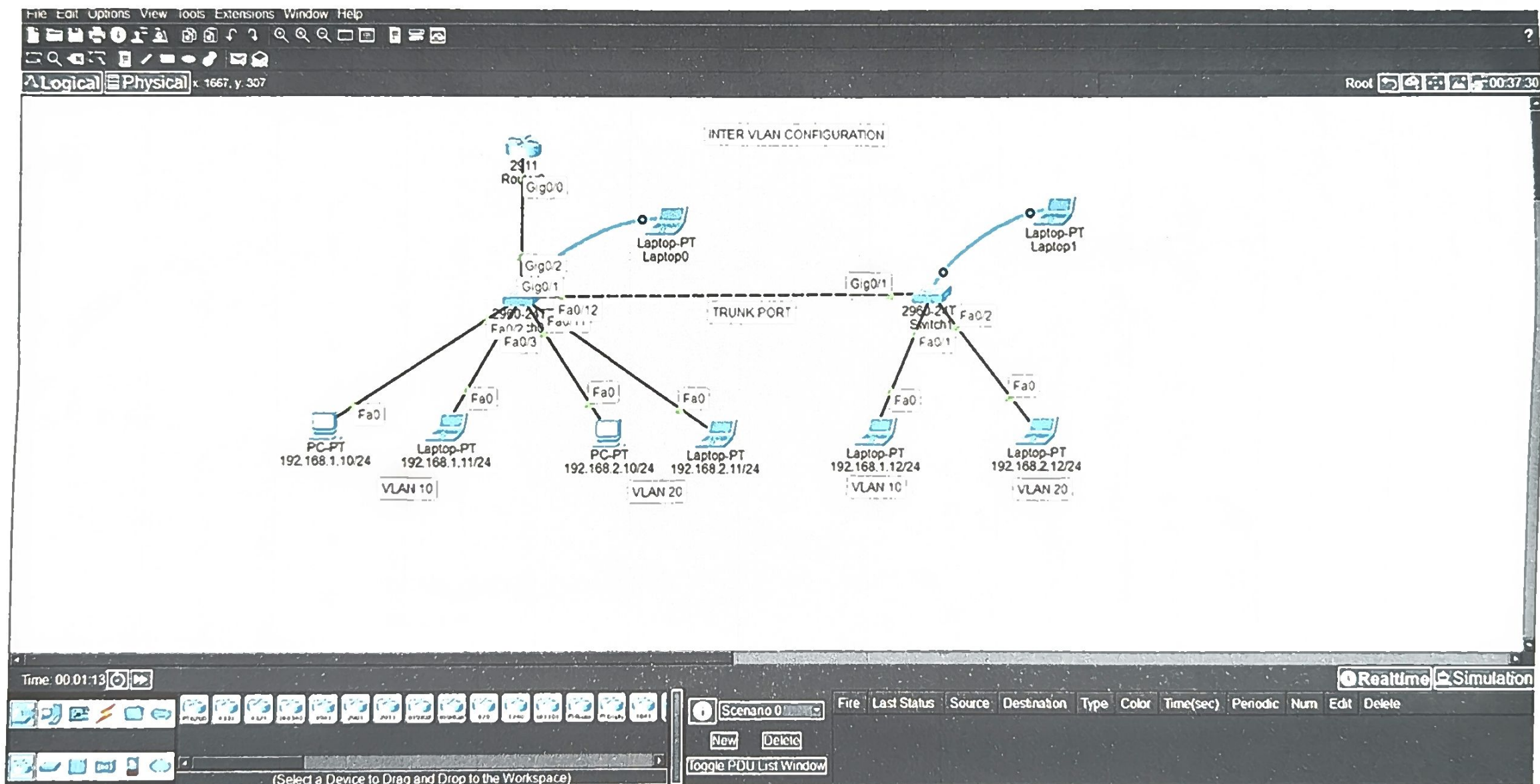
Switch(config)# Name faculties

Switch(config)# Exit

Switch(config)# Interface Range Fa 0/2-3

Switch(config)# Switchport mode access

Switch(config)# Switchport access Vlan 10



```
Switch(config)# Interface range fa 0/11-12
Switch(config)# switchport mode access
Switch(config)# switchport access vlan 20
Switch(config)# Exit
```

```
Switch(config)# Interface gig 0/1
Switch(config)# switchport mode trunk
Switch(config)# switchport trunk allowed all
Switch(config)# Exit
```

```
Switch(config)# Interface gig 0/2
Switch(config)# switchport mode trunk
Switch(config)# switchport trunk allowed all
Switch(config)# Exit
```

• Switch 1:

```
Switch > Enable
Switch# configuration terminal
Switch(config)# vlan 10
Switch(config)# vlan 20
Switch(config)# Interface fa 0/2
Switch(config)# switchport mode access
# switchport access vlan 10
# Interface fa 0/3
# switchport mode access
# switchport access vlan 20
# Exit
```

```
Switch(config)# interface gig 0/1
Switch(config)# switchport mode trunk
Switch(config)# switchport trunk allowed all
Switch(config)# exit
```

• Router 0:

```
Router > Enable
```

```
Router# configuration terminal
```

```
Router(config)# interface gig 0/1
```

```
# no shutdown
```

```
# exit
```

```
Router(config)# interface gig 0/1.10
```

```
# encapsulation dot1q 10
```

```
# IP address 192.168.1.1 255.255.255.0
```

```
# exit
```

```
Router(config)# interface gig 0/1.20
```

```
# encapsulation dot1q 20
```

```
# IP address 192.168.2.1 255.255.255.0
```

```
# exit
```

Verification Protocols:

- To ensure the network is functioning perfectly. The following verification commands and tests must be executed:

- Show vlan brief
- Show Interfaces trunks
- To check end-to-end connectivity from Pc [192.168.1.10] in vlan 10, ~~execute~~ execute an icmp Ping to a host in vlan 20 [192.168.2.10]

C:\> Ping 192.168.2.10